GlucoTwin

An Explainable, Personalized Glucose Digital Twin

Digital Twin Challenge 2026 · Team/institution: [fill in]

Problem & intended use

Glucose trajectories change between visits. A single measurement cannot show short-term behavior.

Goal: fuse historical context with a dynamic glucose stream and forecast near-term values for research review.

Research landscape

Healthcare digital twins are an emerging field. Reviews highlight interoperability, data quality, privacy, ethics and clinical validation.

Existing examples include Twin Health (commercial personalized metabolic health) and academic digital-twin research.

Opportunity & contribution

Integration-focused PoC: auditable virtual state, input-quality gates, forecast history, explanations, uncertainty signaling and reproducible benchmark.

Not a claim that glucose forecasting or digital twins are novel.

Prediction task

Input: previous 12 five-minute glucose readings (60 minutes).

Benchmark target: glucose 30 minutes after input window.

Dashboard forecast: recursive next-step trajectory up to 120 minutes.

No diagnosis or treatment recommendation.

Data and provenance

Included: one fictional profile and 288 synthetic glucose readings.

External candidates: DiaTrend (access/terms must be verified); Synthea for synthetic EHR structure.

No real CGM data downloaded or evaluated.

Architecture and fusion

Historical profile + dynamic glucose stream → ingestion/validation → virtual state → quality gate + forecast → explanation/audit → dashboard.

Historical context is visible in state/UI but not used as model feature in this version.

Dynamic twin workflow

Load SYNTH-001 → inspect source labels → forecast → ingest later reading → state updates and prior forecast clears → generate new forecast.

Invalid/duplicate/out-of-order readings are rejected and audited.

Evaluation (synthetic only)

Patient-separated split: 18 train IDs / 6 held-out IDs; 10,494 train windows / 3,498 test windows.

Persistence: MAE 9.468, RMSE 12.304, R^2 0.434.

HistGradientBoosting: MAE 7.003, RMSE 8.975, R^2 0.6988.

Synthetic data only; not clinical evidence.

Reliability, tests, and limitations

Quality checks: malformed/duplicate/out-of-order inputs, stale stream, low observation count, gaps.

8 automated tests passed in the local environment.

Uncertainty band is heuristic, not calibrated; explanation is descriptive, not causal/SHAP.

Responsible use & next steps

Research/education only; no medical device claims. Do not use for insulin/medication decisions.

Next: authorized real CGM data, verify dataset licensing, calibration on validation data, external evaluation, clinician review, robustness analysis.

Demo & repository

Run: pip install -r requirements.txt; streamlit run app/dashboard.py; pytest -q; python scripts/evaluate.py.

Repository/video/team fields must be completed by team. Demo video plan: README.md (20:30+).